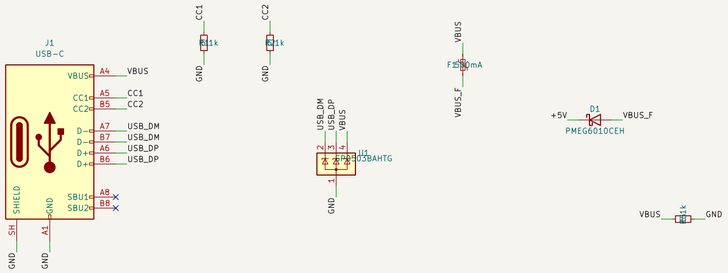
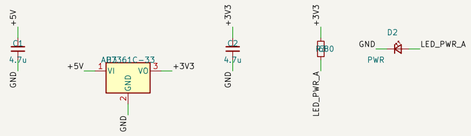
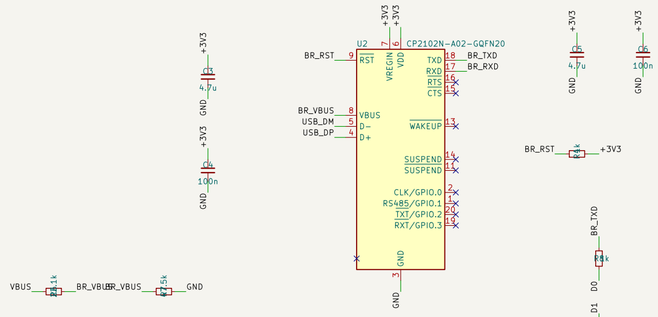
USB-C, ESD, fuse, Schottky, VBUS bleed



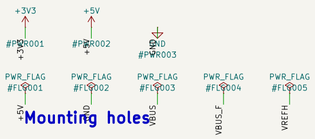
3.3 V LDO, power LED



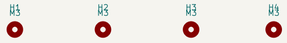
USB-UART: CP2102N on 3.3 V, VBUS sensed (R6, R7) and bled (R5)



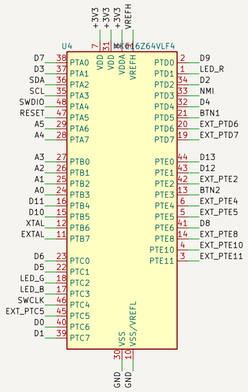
Power nets



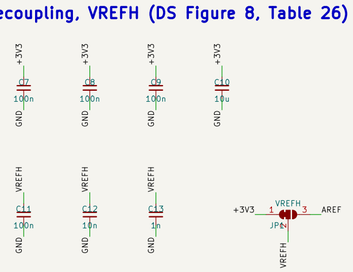
Mounting holes



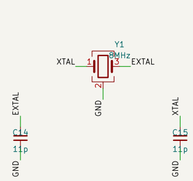
MKE16Z64VLF4



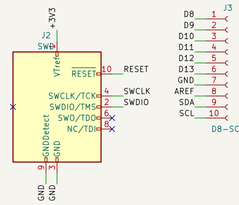
Decoupling, VREFH (DS Figure 8, Table 26)



8 MHz crystal (low-power mode)



SWD (DS Figure 30)



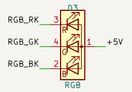
Reset and NMI (DS 6.1.4)

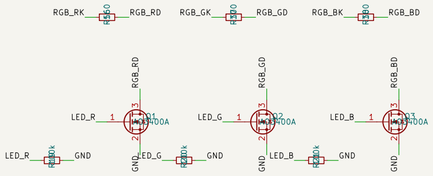


I2C pull-ups



RGB LED, switched from 5 V

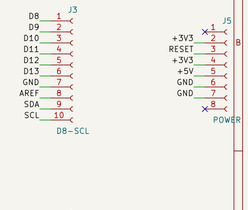




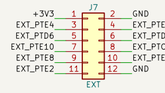
User buttons



Uno R3 shield headers



Extra I/O



EoS KE16Z Board, rev A. Designed by Kartikeydheer Srivastava. Fits Uno R3 shields; 3.3 V I/O.
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Values from NXP KE1xZP48M48SF0 (data sheet) and KE1xZP48M48SF0RM, AN3208, AN1706, ST AN2867;
Silicon Labs CP2102N Rev. 1.5, Diodes DS37274 (AP7361C), Nexperia PMEG6010CEH, Littelfuse, Wurth, AOS.
Firmware: run the 8 MHz oscillator in low-power mode (HGO = 0); high-gain mode would
overdrive the crystal (AN3208 Eqn. 1), and RF must not be fitted (DS Table 40 note 3).

Power: USB-C VBUS through F1 and D1 to the 5 V rail (LDO, RGB LED, 5V pin);
D1 keeps a shield's 5 V off the host. The CP2102N runs from 3.3 V and senses VBUS
(R6, R7); R5 bleeds D1's leakage off VBUS, so it connects only while a host is present.

Uno R3 mounting holes (4 x 3.20 mm) and pin order: Arduino UNO R3 product reference manual (SKU
Uno R3 outline (2700 x 2100 mil) and shield header positions: Matthew Beckler, "Arduino Uno R3" di
Generated by tools/eos_boards from the board's design file
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Title: EoS KE16Z Board

Size: A2	Date: 2026-10-12	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1